

KHMELNYSKYI, Ukraine

WMO: 334290

Lat: 49.3536N

Lon: 26.9375E

Elev: 1148

StdP: 14.10

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.2	5.6	-4.9	4.4	0.9	0.7	5.9	6.1	26.9	31.2	23.8	30.1	7.3	60	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	85.2	68.0	82.2	66.6	79.4	65.4	70.3	81.3	68.6	78.7	67.1	76.4	8.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	102.3	75.2	65.1	96.8	73.4	63.6	91.9	71.9	34.9	81.6	33.5	78.7	32.2	76.5	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.9	20.1	18.0	DB	-5.5	89.4	6.8	2.6	-10.4	91.3	-14.4	92.8	-18.2	94.3	-23.1	96.2	(4)
(5)			WB	-5.9	73.6	6.7	2.9	-10.7	75.7	-14.6	77.4	-18.4	79.1	-23.2	81.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	24.4	27.1	35.4	48.4	58.1	63.8	67.2	66.3	57.4	46.8	37.1	27.7	(6)	
	DBStd	17.19	9.73	9.87	8.53	7.84	7.11	6.01	5.53	5.87	6.99	7.70	8.41	9.26	(7)	
	HDD50	3276	793	640	455	118	15	0	0	0	16	154	393	692	(8)	
	HDD65	6958	1258	1060	917	498	233	94	41	54	244	564	838	1157	(9)	
	CDD50	2089	0	0	2	71	266	414	533	506	237	55	5	0	(10)	
	CDD65	297	0	0	0	1	19	58	109	95	15	0	0	0	(11)	
	CDH74	2321	0	0	0	12	167	401	822	783	134	2	0	0	(12)	
	CDH80	525	0	0	0	1	18	59	204	220	23	0	0	0	(13)	
(14) Wind	WSAvg	9.0	10.0	10.3	10.4	9.6	8.6	8.0	7.7	7.4	8.1	8.6	9.9	9.7	(14)	
(15) Precipitation	PrecAvg	25.2	1.3	1.4	1.4	1.9	2.2	3.5	3.9	2.6	2.4	1.7	1.7	1.5	(15)	
	PrecMax	34.6	2.3	2.4	3.5	5.6	4.6	7.4	9.7	8.2	6.2	5.4	3.4	3.9	(16)	
	PrecMin	16.1	0.3	0.4	0.4	0.4	0.7	1.3	1.3	0.0	0.3	0.0	0.1	0.6	(17)	
	PrecStd	4.2	0.5	0.6	0.8	1.0	0.9	1.5	2.3	1.8	1.6	1.1	0.7	0.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	50.2	60.9	74.5	82.6	84.7	89.0	89.8	83.1	72.2	60.2	48.6	(19)	
		MCWB	41.4	44.0	48.7	57.4	64.1	68.1	70.2	68.3	64.5	59.2	52.8	46.1	(20)	
	2%	DB	40.6	44.8	55.4	69.2	78.6	81.6	85.3	85.4	77.7	66.6	54.5	44.8	(21)	
		MCWB	38.4	41.3	46.2	54.7	61.7	66.7	68.9	67.7	61.5	56.9	50.4	42.9	(22)	
	5%	DB	38.1	41.3	51.0	65.2	74.9	78.8	81.8	82.0	73.7	61.7	51.2	41.2	(23)	
		MCWB	36.3	38.8	43.6	52.5	60.9	65.4	67.6	66.2	60.1	54.5	48.4	39.7	(24)	
	10%	DB	35.6	38.3	46.9	61.3	71.1	75.7	78.6	78.3	69.2	57.7	48.3	38.2	(25)	
		MCWB	34.2	36.2	41.5	50.4	58.9	64.2	66.4	64.9	58.7	52.4	46.2	36.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.6	45.2	50.4	58.8	67.1	71.5	72.4	72.3	66.5	61.3	53.7	46.8	(27)	
		MCDB	43.9	49.2	58.8	72.4	77.7	81.1	83.3	84.2	78.8	68.8	58.0	48.3	(28)	
	2%	WB	38.8	41.7	47.1	56.1	64.1	68.9	70.5	69.8	63.6	57.7	51.2	42.8	(29)	
		MCDB	40.2	44.4	53.8	66.5	74.4	78.1	81.8	81.4	73.4	64.1	54.1	44.4	(30)	
	5%	WB	36.5	39.0	44.5	53.9	62.0	67.1	69.0	67.9	61.5	55.4	48.8	39.9	(31)	
		MCDB	37.8	41.1	49.3	62.9	71.9	76.1	79.4	77.6	70.4	60.9	51.0	41.2	(32)	
	10%	WB	34.4	36.4	42.1	51.7	60.0	65.3	67.4	66.1	59.8	53.0	46.3	37.0	(33)	
		MCDB	35.4	38.2	46.3	59.4	69.1	73.6	76.5	75.6	67.5	57.0	48.3	38.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	9.0	11.8	15.7	17.2	16.4	17.0	17.7	15.5	12.9	8.5	7.5	(35)	
		MCDBR	8.0	12.2	18.5	21.4	22.3	20.8	21.8	22.8	22.5	19.1	14.1	9.4	(36)	
	5% WB	MCWBR	6.6	9.1	11.1	10.1	10.0	9.5	8.9	9.0	9.7	10.3	9.4	7.7	(37)	
		MCWBR	7.6	11.3	16.8	19.5	19.7	18.3	19.3	20.1	19.2	16.8	12.5	8.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.306	0.337	0.375	0.429	0.403	0.402	0.433	0.432	0.393	0.360	0.337	0.304	(40)	
		taud	2.237	2.190	2.200	2.124	2.270	2.291	2.219	2.215	2.284	2.359	2.336	2.305	(41)	
	Ebn at Noon		230	250	261	258	272	272	261	255	253	241	218	215	(42)	
		Edh at Noon	24	32	37	45	41	40	43	41	34	27	22	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		300	567	943	1375	1760	1864	1790	1578	1102	675	290	209	(44)	
	RadStd		56	89	130	146	154	140	129	113	168	99	32	37	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.33	N/A	N/A	+1.64	+0.92	N/A	N/A	-342	+247	+95	(46)
(47)	Regional (13 neighbors)	+1.26	N/A	N/A	+1.71	+0.90	+0.79	-250	-378	+227	+94	(47)

Nomenclature: See separate page